Explainable AI-Enhanced Detection of Cyber Attacks on AGC Systems with Varying RES Penetration

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***Abstract*—This study addresses the limitations of traditional methods in detecting False Data Injection Attacks (FDIA) on Automatic Generation Control (AGC) systems, particularly due to State Estimation (SE) algorithms' tolerance for measurement errors. To overcome these limitations, six machine learning algorithms are compared across different levels of Renewable Energy Source (RES) penetration. As RES levels increase, the impact of cyber-attacks becomes more pronounced, with Convolutional Neural Networks (CNN) maintaining high performance, followed closely by the Bagged Trees (BT), algorithm as the second most successful.**

Additionally, Explainable Artificial Intelligence (XAI) techniques, such as Shapley Additive Explanations (SHAP), are used to clarify the role of critical features, like frequency and Area Control Error (ACE), in attack detection, making the model's decision-making process more transparent for system operators. The AGC model was implemented in MATLAB/Simulink within a two-area, four-machine power system, using realistic turbine and speed governor models rather than linearized approximations. Results show that CNN and BT are highly effective in FDIA detection within high RES environments, indicating their reliability for enhancing AGC system security under realistic operational conditions. Furthermore, the interpretability provided by SHAP assists operators in understanding how critical measurements affect detection, thus building confidence in Artificial Intelligence (AI) driven protective mechanisms for modern power grids.

Keywords—Automatic Generation Control, Bagged Trees, Explainable Artificial Intelligence, Renewable Energy

I. Introduction

AGC is a critical mechanism used in multi-area power systems to maintain system frequency and manage tie-line power flows. Specifically, AGC monitors the ACE, which reflects both frequency deviation (i.e., how far the system frequency is from its nominal value) and tie-line power exchange deviation between interconnected areas. By adjusting generator setpoints, the AGC ensures that each area's frequency remains close to the nominal level (e.g., 50 or 60 Hz) and that scheduled power transfers across tie-lines are upheld. The frequency and tie-line measurements obtained by the sensors are transmitted to the control center of each area via the Distributed Network Protocol (DNP3), which has vulnerabilities against cyber-attacks [1] - [2]. Attackers who use these vulnerabilities can inject false data into the system that will affect the operation of the AGC. Recent studies demonstrate that optimal false data injection attacks (OFDIAs) on AGC systems can bypass conventional bad data detection mechanisms, leading to significant consequences such as frequency instability or triggering protection mechanisms like load shedding relays. These attacks leverage precise manipulation of tie-line power and generator frequency measurements, highlighting vulnerabilities in existing control frameworks [3]. Although there are many studies examining the effects of cyber-attacks on the AGC system, the exploration of detection methods in the presence of high levels of RES remains limited.

Integration of RES into the electrical grid is increasing to reduce carbon emissions and dependency on finite resources. Yet, high RES penetration introduces generation intermittency and reduces overall system inertia, thereby affecting frequency stability [4]. Recent studies have highlighted that fluctuations caused by wind power and the resulting reduction in system inertia provide attackers with opportunities to amplify FDIA effects on AGC stability [5]. When integrated into AGC systems, these low-inertia resources make the grid more vulnerable to cyber-attacks, as fluctuations in frequency are magnified [4] [6] [7].

The Random Forest algorithm has demonstrated high accuracy in detecting both single-point and multi-point FDIA, adapting effectively to new attack patterns as they emerge. However, its performance decreases when all AGC system parameters are simultaneously attacked, a scenario with low practical probability but significant complexity [8].

In [9] Sparse Logistic Regression, k-Nearest Neighbors (k-NN), Multilayer Perceptron (MLP), AdaBoost, and Support Vector Machine (SVM) were applied to anomaly detection in a power system context. Likewise, [10] employed a Recurrent Neural Network (RNN)-based LSTM architecture for cyber-attack detection, achieving over 90% accuracy. Despite these methods, the synergy of CNN and BT in detecting FDI attacks under varying RES penetration has not been fully explored, especially using realistic turbine and governor models in place of linearized approximations.

In another study, three classification techniques were tested using various Heuristic Feature Selection Techniques [11]. The authors found that the SVM and k-NN algorithms produce a more accurate result than the neural network, but these methods are not applicable for large systems.

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Reference [7] describes various and hybrid FDI attacks and further analyzes them on multiarea AGC. A data-based method consisting of fuzzy logic and neural networks is presented to detect attacks. In addition, by integrating large-scale renewable energy sources into the AGC system, three simulation scenarios were created according to the situation with and without renewable energy generation, and the performance of detecting FDIs was verified.

Cyber-attacks on AGC systems have been examined in many studies, but it has been observed that studies integrating renewable energy sources are insufficient. In addition, the selection criteria of the algorithms used to detect cyber-attacks on RES integrated AGC systems are a complete black box. Finally, in this study, which features contribute to the proposed algorithm in label selection and how are explained using XAI methods.

II. System Description and Model

The main purpose of AGC is to control the deviation in system frequency and bring it close to zero. Frequency deviation $\Delta f(t)$ is defined as in (1),

$$\Delta f(t) = f_{ref} - f(t) \tag{1}$$

The reference frequency f_{ref} represents the desired system frequency, while $f(t)$ denotes the actual frequency at time t. Any deviation prompts the AGC to generate a control signal ($u(t)$) to correct the frequency imbalance by adjusting the generator output power (P_{gen}) through the turbine and governor system. In multi-area AGC systems, each area independently manages its frequency and power variations while also controlling power exchanges with interconnected areas.

A. Multi-Area AGC Model

The frequency dynamics adapted from the [12], for each area can be formulated as in (2),

$$\frac{d\Delta f_i(t)}{dt} = \frac{1}{2H_i}(\Delta P_{m,i}(t) - D_i \Delta f_i - \Delta P_{tie,i}(t)). \tag{2}$$

$\Delta f_i(t)$ is the frequency deviation of the i-th area, H (s) is the inertia constant of the generators, D (pu) is the load damping coefficient, $\Delta P_{m,i}(t)$ (pu) is the mechanical power change of the i-th area, $\Delta P_{tie,i}(t)$ (pu) is the power exchange between two areas. The tie-line power exchange between two fields is expressed as in (3),

$$\Delta P_{tie,1}(t) = T_{12}(\Delta f_1(t) - \Delta f_2(t)). \tag{3}$$

T_{12} is the constant expressing the synchronized power coefficient. In a multi-area AGC system, the ACE equations are used for each area. ACE for each area depends on the frequency deviation of that area and the power exchange difference with other areas. ACE is defined as in (4),

$$ACE_i(t) = \Delta P_{tie,i}(t) + \beta_i \cdot \Delta f_i(t). \tag{4}$$

$\Delta P_{tie,i}(t)$ (pu) represents the tie-line power exchange change for the i-th area with other areas, β_i (in pu/Hz) is the frequency bias coefficient for that area, and $\Delta f_i(t)$ (pu) denotes the frequency deviation in that area. The AGC system tries to minimize the ACE value of each area. For each area, the control signal is

$$u_i(t) = -K_i \cdot ACE_i(t). \tag{5}$$

K_i is the control gain for that area.

B. Governor and Turbine System

Turbine systems convert different types of energy from energy sources into mechanical energy. These energy sources can be water, fossil fuel, nuclear or wind. Unlike many works that rely on linearized turbine-governor approximations, this study uses a tandem compound double reheater steam turbine model to capture more accurate dynamics. The equations for each turbine stage are provided in (6)–(9). The signal $u_i(t)$ generated by the AGC is transmitted to the turbine and governor systems. The turbine and governor dynamics adjust the generator output power in response to this control signal from the AGC. The state equations for each turbine stage are derived using the $IEEEG_1$ model given in [13] and are as follows,

$$\frac{d\Delta P_{VHP,i}(t)}{dt} = \frac{1}{T_4}(K_4 \cdot u_i(t) - \Delta P_{VHP,i}(t)) \tag{6}$$

$$\frac{d\Delta P_{HP,i}(t)}{dt} = \frac{1}{T_3}(K_3 \cdot \Delta P_{VHP,i} - \Delta P_{HP,i}(t)) \tag{7}$$

$$\frac{d\Delta P_{IP,i}(t)}{dt} = \frac{1}{T_2}(K_2 \cdot \Delta P_{HP,i} - \Delta P_{IP,i}(t)) \tag{8}$$

$$\frac{d\Delta P_{LP,i}(t)}{dt} = \frac{1}{T_1}(K_1 \cdot \Delta P_{IP,i} - \Delta P_{LP,i}(t)) \tag{9}$$

where T_4 (s) is the steam chest time constant, T_3 (s) is the reheater time constant T_2 (s) is the second reheater time constant and T_1 (s) is the crossover time constant. K_4 is the very high pressure turbine power fraction, K_3 is the high pressure turbine power fraction, K_2 is the medium pressure turbine power fraction and K_1 is the low pressure turbine power fraction.

C. Speed-Governing System

This study utilizes electro-hydraulic speed control systems to regulate steam turbine speed. The turbine governor manages speed and power output based on speed feedback and reference power (P_{ref}) signals. The speed feedback gain K_G and the valve control gain K_P re key parameters, with the speed reference (ω_{ref}) and feedback ($\omega \cdot K_G$) relationships expressed in (10),

$$\Delta\Omega = \omega_{ref} - (\omega \cdot K_G). \tag{10}$$

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Speed deviation signal ($\Delta\Omega$) is the basic input for determining valve opening and other control signals. The servomotor controls the opening of the valves, and the valve control signal (C_v) is modeled in (11),

$$\frac{dC_v}{dt} = \frac{1}{T_{SM}}(u_{servo} - C_v) \tag{11}$$

here, T_{SM} epresents the servomotor time constant, and u_{servo} is the valve opening command. This differential equation models how the valve position (C_v) evolves over time in response to u_{servo}. The speed deviation and servomotor dynamics are based on established methodologies in steam turbine control systems, as detailed in [13] [14].

Fig. 1 compares frequency responses of linear and nonlinear models after a 0.1 pu load increase in Area-1 at t=0, highlighting significant differences due to system nonlinearities. Nonlinear models account for effects like valve saturation, steam chest dynamics, and reheater behavior, which linear models often neglect. Using realistic models enhances simulation accuracy, providing more reliable insights for evaluating FDIA impacts and improving detection strategies.

III. Simulation Results

A. Dataset Generation

Machine learning algorithms require labeled datasets for training and testing. This study defines four classes: attacks on Area-1 frequency, Area-2 frequency, tie-line power, and non-attacked cases. Scenarios were designed to simulate realistic conditions, incorporating load variations, RES penetration rates, and common cyber-attack patterns, capturing grid dynamics under normal and attack scenarios. Fig. 2 is the test system used in this simulation.

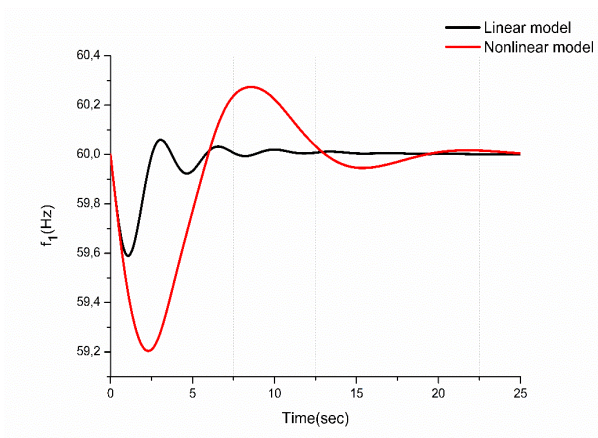

Fig. 1. Area-1 frequency response according to linear and nonlinear models.

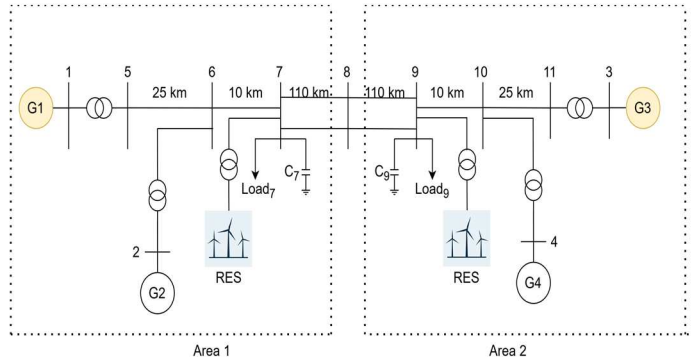

Fig. 2. Two Area four machine test system with RES integration *[14]*.

As seen in Fig. 2 RES units are connected at buses 7 and 9, while AGC is integrated with generators G1 and G3 (highlighted in yellow). The remaining generators are managed by primary control loops, reflecting their exclusion from AGC. This configuration illustrates the interaction between RES and AGC systems under varying RES penetration levels, which is critical for analyzing system stability and intrusion detection.

The RES penetration levels of 0%, 10%, 20%, 30%, and 50% correspond to typical operational scenarios ranging from low to high renewable integration in modern power grids. These levels aim to simulate varying system inertia and their impact on vulnerability to FDI attacks.

For each RES integration level, a separate dataset is created by running the MATLAB/Simulink model. For each dataset, there are a total of 11 features, namely the frequency (f_{m1}, f_{m2}, f_{m3}, f_{m4}) and power values (P_{m1}, P_{m2}, P_{m3}, P_{m4}) of the four generators shown in Fig. 2, ACE_1, ACE_2 and tie-line power (P_{tie}) values. The algorithms trained using these features predict four different classes as attack on Area-1 frequency, attack on Area-2 frequency, attack on tie-line power values and nonattacked situations. Each dataset consists of approximately 90,000 rows of samples with an approximately equal distribution among the four classes. This balanced distribution increases the generalization capabilities of the models by ensuring that they are trained without bias towards a particular class. All parameter values required for the simulations in this study are taken from [15].

B. Classifier Structure

AGC maintains grid frequency by regulating power plant set points, but including all measurements in the classifier can lead to high dimensionality and overfitting. To address this, the Bagged Trees (BT) algorithm, derived from "bootstrap aggregating" by Leo Breiman [16], is used. BT combines multiple decision trees and makes final decisions through majority voting, offering faster training and prediction via parallel execution, making it scalable for large dataset. The BT algorithm follows the following steps:

- The given data set $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ is resampled by bootstrap method and data sets D_1, D_2, …, D_B are created.
- A decision tree T_i is trained on each D_i data set. These trees are used to classify the data according to features.

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- Each decision tree T_i produces a class prediction $\hat{y}_i$ for an input data x.
- The final classification decision is based on the voting results of trees $T_1, T_2, \ldots, T_m$.

$$\hat{y}_{bagging} = mode(\{\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_m\}) \tag{12}$$

Equation (12) describes the selection of the most predicted class label as the final class label.

IV. Results

A. Results of Attacks on the AGC System

AGC adjusts power plant setpoints to maintain the desired network frequency based on sensor measurements sent via DNP3. However, DNP3's vulnerability to cyber-attacks allows attackers to manipulate transmitted data [1] [17] [2]. This can lead to incorrect ACE calculations, causing improper generator adjustments and unwanted frequency changes in the grid. The attacks, including ramp, scaling, compensation, and combined attack, were applied with different parameter values at three different points; f_1, f_2, p_{tie}.

1) Scaling Attack: Scaling attack is the modification of signals by multiplying them by a certain value according to the attack parameter λ_s.

2) Ramp Attack: Ramp attack is the process of changing actual measurements by adding a ramp function $\lambda_r * t$, which increases or decreases proportionally with time, to the correct measurements.

3) Compensation Attack: The attack model here presents the attack model examined in the article [18]. The amount of deviation from the nominal value of the frequency and tie-line power during the attack is multiplied by the coefficient m before it is sent to the control center. Therefore, the value of ACE_i changes to $m_i \; x \; ACE_i$.

Positive compensation attacks slow the AGC system's adjustment speed, extending fluctuation periods without fully collapsing state variables. These attacks cause frequent ACE signal fluctuations, making them easier to detect. However, if the ACE signal exceeds its 0.05 per unit limit, repeated attacks can severely degrade system frequency. Negative compensation attacks work against AGC adjustments, significantly increasing instability during load fluctuations. In contrast, scaling attacks are generally ineffective when the load remains constant.

Fig. 3 illustrates the effects of compensation attacks with different m_1 parameters on Area 1 frequency. When $m_1 = -1$ (blue line), the load-shedding frequency of 59.3 Hz [19] is reached fastest, but as shown in Fig. 4, ACE_1 (pu) quickly exceeds ± 0.05 pu. In contrast, with $m_1 = -0.05$ (red line), ACE_1 remains within limits as the system frequency approaches the load-shedding threshold. Positive compensation attacks, as shown in Fig. 3, fail to significantly alter the system frequency. Attack parameters were consistently chosen across scenarios to adhere to defined constraints.

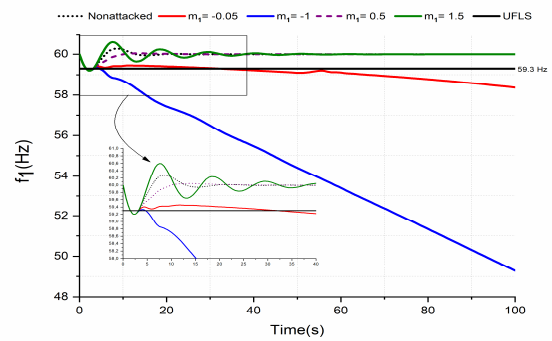

Fig. 3. System frequency during compensation attacks.

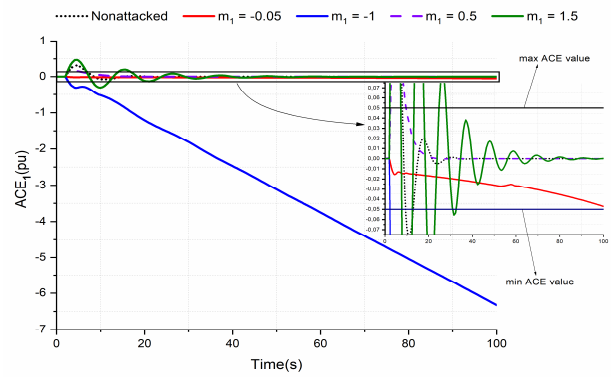

Fig. 4. ACE signal of Area 1 during compensation attacks.

B. Attack Detection Performance of Algorithms for Different RES Entegration Rate

To examine the advantages and disadvantages of machine learning methods, datasets under various attack scenarios are trained to obtain Decision Tree (DT), Long-Short Term Memory (LSTM), k-NN, MLP, CNN, BT detection models.

Fig. 5 and Fig. 6 display the confusion matrices of the BT Algorithm for systems without RES integration and with 50% RES integration, respectively. The results indicate that RES integration negatively impacts the BT Algorithm's intrusion detection performance. This trend is consistent across all machine learning algorithms tested in this study, with detection performance decreasing as the RES ratio increases. Table I summarizes the performance metrics for the algorithms under various RES integration levels.

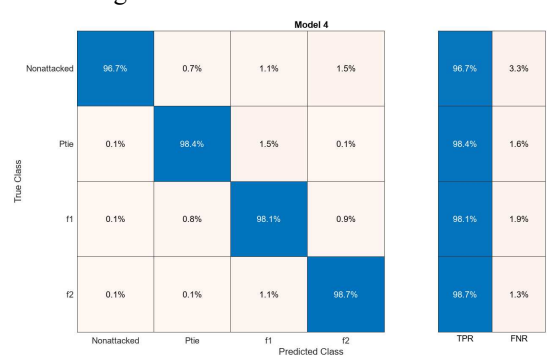

Fig. 5. BT Algorithm intrusion detection confusion matrix. (0% RES)

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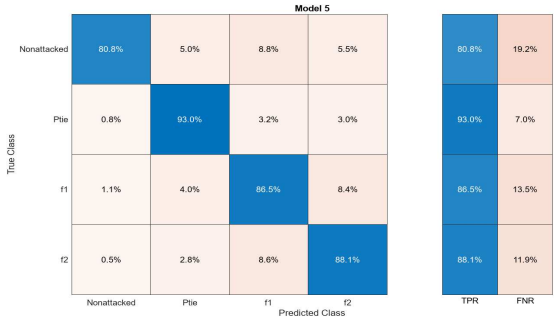

Fig. 6. BT Algorithm intrusion detection confusion matrix. (50% RES)

1) Robustness of CNN and BT: CNN excels due to its ability to learn complex temporal and spatial features from the data, capturing non-linear relationships in frequency, ACE, and tie-line power. BT combines multiple decision trees to form a stronger classifier via bootstrap aggregating, which reduces variance and mitigates overfitting. This ensemble nature provides robustness against the increased variability introduced by higher RES penetration.

2) Impact of RES Penetration: Higher RES levels introduce faster frequency dynamics and lower system inertia, making the grid more vulnerable to FDIA. CNN's convolutional filters adapt to these rapid changes in frequency signals, whereas BT's ensemble strategy effectively handles higher variance in measurement data.

3) Reliability of Detection Methods: The proposed FDIA detection methods, CNN and BT, generalize well due to diverse training datasets covering various attack types and RES levels, reducing overfitting. CNN captures complex time-series patterns, while BT provides robust classification through decision tree aggregation. These methods enhance detection of unseen attack intensities and ensure reliability under unmodeled conditions with multi-metric assessments and redundant sensor data.

C. *Practical Integration of CNN and Bagged Trees into AGC for Real-Time FDIA Detection*

By leveraging parallel processing, existing data channels, and XAI tools, CNN and BT are both feasible and beneficial for real-time FDIA detection within modern AGC systems, thereby enhancing power grid resilience against cyber threats.

1) Scalability and Parallelization: CNN inference can be accelerated using Graphics Processing Unit (GPU) *[20]* or multi-core Central Processing Unit (CPU) resources, which are increasingly available in modern control centers. BT relies on aggregating multiple decision trees, which can be trained and run in parallel on standard CPU architectures. This enables fast classification even under high data throughput.

2) Data Flows: Both methods require continuous data streams already gathered by Supervisory Control and Data Acquisition (SCADA) or Phasor Measurement Unit (PMU). Existing communication protocols (e.g., IEC 61850, DNP3) can feed these algorithms with minimal modification to current infrastructure [21].

Table I. Performance evaluation of the algorithms.

Algorithm	0% RES			50% RES		
	Precision	Recall	F1 Score	Precision	Recall	F1 Score
CNN	0.979	0.976	0.977	0.973	0.966	0.969
BT	0.984	0.980	0.982	0.894	0.879	0.886
LSTM	0.959	0.948	0.953	0.758	0.711	0.734
MLP	0.979	0.975	0.977	0.885	0.863	0.871
k-NN	0.881	0.863	0.869	0.707	0.653	0.673
DT	0.888	0.860	0.870	0.683	0.605	0.622
AdaBoost	0.832	0.818	0.821	0.635	0.542	0.554

V. Understanding Detection Mechanisms: XAI in the Context of FDI Attacks on AGC Systems

Machine learning models for grid security often face trust issues due to their "black box" nature. XAI methods improve transparency and reliability by making model outputs understandable, aiding in FDI attack detection on AGC systems.of AI-based solutions for detecting FDI attacks on AGC systems. SHAP quantifies the contribution of each feature to the prediction of a model to explain the output of any machine learning model by calculating the marginal contribution of that feature in all possible subsets.

Highlighting features with the highest SHAP values helps operators identify suspicious deviations. For example, large SHAP values in certain generator frequencies suggest potential data tampering in those channels, guiding operators to inspect sensors, SCADA channels, or local controllers linked to these features, thus streamlining root cause analysis. SHAP analysis can reveal targeted attacks. Operators can increase monitoring of the targeted channel, adjust anomaly thresholds for sensors with repeated suspicious activity.

The SHAP summary graph (Fig. 7) from the BT model with 50% RES integration highlights key features influencing decisions under high RES conditions. The most critical features for detecting attacks are ACE_2 and ACE_1 which reflect the AGC system's ability to manage power and frequency imbalances. The greater prominence of ACE_2 indicates that control dynamics in the second area play a more significant role in attack detection, demonstrating the model's sensitivity to imbalances in this area.

Fig. 7 highlights the importance of generator frequencies, with (f_{m1}) and (f_{m3}), actively controlled by the AGC, identified as strong attack indicators. These frequencies provide critical signals as the AGC adjusts them to maintain balance. In contrast f_{m2} and f_{m4}, outside AGC control, have limited utility in attack detection.

Since these generators are not directly regulated by the AGC system, their ability to signal imbalances is restricted. Tie-line power flow is a significant feature after ACE values and AGC-integrated generator frequencies, providing a reliable indicator of attack scenarios by detecting deviations in inter-area power flow.

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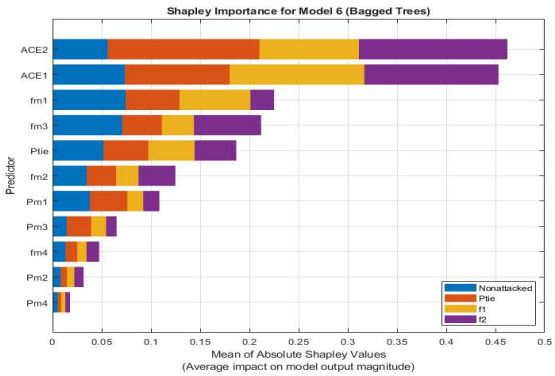

Fig. 7. SHAP summary graph for the BT algorithm model 50% RES integration.

Generators two and four, excluded from AGC control, have minimal impact on attack detection, as shown in the SHAP summary graph. SHAP enhances the BT model's transparency, highlighting key features like ACE and ptie, fostering trust and improving real-time decision-making, particularly in high RES scenarios.

VI. Conclusion

This study evaluated six machine learning algorithms for detecting FDI attacks on AGC systems under varying RES penetration levels. CNN demonstrated the highest resilience under high RES conditions, while BT offered a strong balance between accuracy and computational efficiency. Simpler models struggled under variability, emphasizing the importance of tailored algorithms for grid conditions.

The integration of XAI techniques further enhances the applicability of these models by providing transparency into decision-making processes. Features such as ACE and tie-line power flow were identified as critical indicators, enabling grid operators to better understand and trust AI-driven solutions.

Future research should focus on hybrid approaches that combine CNN's robustness with BT's efficiency to address diverse operational requirements. Incorporating real-time data streams and advanced feature engineering could also enhance adaptability and resilience against emerging cyber threats in modern power grids.

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